



**R.B.INSTITUTE OF MANAGEMENT
STUDIES(RBIMS) MCA SEMESTER-3**

SUBJECT NAME: : Design and Analysis of Algorithms

SUBJECT CODE: : MC03094011

Assignment-1

Issue Date:12/08/2026 Submission Date:18/08/2026 Short Questions:

1. Trace the execution of Insertion Sort for the following array:
45, 23, 67, 12, 89, 34, 10
2. Arrange the following functions in increasing order of growth:
 $1, \log n, n, n \log n, n^2, n^3, 2^n, n!$
3. Define an algorithm.
4. Define time complexity.
5. Define space complexity.
6. Write the best, average, and worst-case time complexity of Insertion Sort.

Long Questions:

1. Explain the role of algorithms in computing with suitable examples.
2. Explain algorithms as a technology.
3. Describe the characteristics of a good algorithm.
4. Explain the steps involved in designing an algorithm.
5. Explain Insertion Sort with a suitable example.
6. Explain asymptotic notation and its importance.
7. Describe between Big-O, Big- Ω , and Big- Θ notation.
8. Explain standard notations and common mathematical functions used in algorithm analysis.